



car-dealership-eks

Star Fork Watch Private Forks

71 files main branch Python 38% JavaScript 35% HCL 18% Shell 9% MIT License

Complete Repository Structure

car-dealership-eks/	Root repository
.gitignore	Blocks secrets, tfstate, .env, node_modules
README.md	Full deployment guide & architecture docs
docker-compose.yml	Local dev stack (Postgres, Redis, Backend, Frontend)
.github/	GitHub Actions workflows
workflows/	4 CI/CD pipeline definitions
terraform.yml	Terraform plan & apply on infra/ changes
build-push.yml	Docker build + Trivy scan + ECR push
deploy.yml	EKS deploy via kubectl + Helm upgrades
security-scan.yml	Weekly Trivy, tfsec, Checkov, GitLeaks
terraform/	All AWS infrastructure as code
providers.tf	AWS, Kubernetes, Helm providers + S3 backend
main.tf	Root module — wires all modules together
variables.tf	All input variables with defaults
outputs.tf	Cluster name, ECR URLs, RDS/Redis endpoints
terraform.tfvars.example	Safe reference config (real tfvars gitignored)
modules/	8 reusable Terraform modules
vpc/	VPC, subnets (public/private/db), NAT GWs, routes
eks/	EKS cluster, node group, OIDC, KMS, add-ons
rds/	PostgreSQL 16, Multi-AZ, backups, parameter group
elasticache/	Redis 7 replication group, TLS, Multi-AZ
ecr/	Private repos, immutable tags, scan, lifecycle
iam/	Cluster role, node role, IRSA pod role
security-groups/	EKS cluster/node/RDS/Redis SG rules
cloudfront/	CDN distribution, S3 logs, TTL policies
frontend/	Next.js 14 application
Dockerfile	3-stage build: deps → builder → runner (non-root)
.dockerignore	Excludes node_modules, .next, .env files
package.json	Next.js 14, React 18, Tailwind CSS dependencies
next.config.js	Standalone output, CSP headers, backend URL env
tailwind.config.js	Brand colours, Inter font, fade-in animation
postcss.config.js	Tailwind + Autoprefixer build pipeline
src/	All Next.js source code
pages/	Next.js pages + API proxy routes
index.jsx	Homepage: SSR hero, filters, car grid, contact
_app.jsx	Global layout: Navbar + Footer wrapper
_document.jsx	HTML shell: Inter font, lang=en, antialiased
api/	BFF proxy routes (browser never hits backend direct)
cars.js	Proxies GET /api/cars with 1-min cache header
inquiries.js	Proxies POST /api/inquiries contact form
health.js	Liveness probe endpoint → {"status":"ok"}
components/	Reusable React components
CarCard.jsx	Vehicle card: image, specs, price, inquiry modal
HeroSection.jsx	Gradient hero, quick search, trust badges
SearchFilters.jsx	Make/model/year/price filter bar + Search btn
ContactForm.jsx	Validated form with loading/success/error states
Navbar.jsx	Sticky nav, logo, links, mobile hamburger menu
Footer.jsx	3-column footer: brand, links, contact info
styles/	Global CSS
globals.css	Tailwind directives + .btn-primary / .input-field

backend/	FastAPI Python application
Dockerfile	2-stage: gcc/psycpg2 build → slim runner, non-root uid 1001
.dockerignore	Excludes __pycache__, tests, .venv, migrations from image
requirements.txt	FastAPI, SQLAlchemy, AsyncPG, Redis, Prometheus, Pydantic v2
alembic.ini	DB migration config — reads DB_* env vars at runtime
app/	FastAPI application package
__init__.py	Package marker
main.py	App bootstrap: lifespan, CORS, Prometheus, request logging
database.py	Async SQLAlchemy engine from DB_* env (never hardcoded)
cache.py	Redis async pool, cache_get/cache_set helpers, 5-min TTL
models/	SQLAlchemy ORM models
__init__.py	Car model (make/model/year/price/vin...) + Inquiry model
routes/	FastAPI route handlers
__init__.py	Package marker
cars.py	GET /api/cars (Redis cache → PostgreSQL) + GET /{id}
inquiries.py	POST /api/inquiries — validates + inserts to DB
health.py	GET /health (liveness) + GET /ready (readiness probe)
migrations/	Alembic async migrations
env.py	Async migration runner, reads DB_* env at runtime
versions/	Versioned migration scripts
001_initial.py	Creates cars table + inquiries table with indexes

k8s/	All Kubernetes manifests — applied by GitHub Actions
namespaces/	Kubernetes namespace definitions
namespaces.yaml	frontend, backend, monitoring, external-secrets (Istio labelled)
deployments/	Application workload definitions
frontend-deployment.yaml	Deployment + Service + SA — 3 replicas, zone spread, probes
backend-deployment.yaml	Deployment + Service + SA (IRSA annotated) — read-only FS
redis-dev.yaml	In-cluster Redis (dev/staging only — prod uses ElasticCache)
network-policies/	Zero-trust pod networking rules
network-policies.yaml	Default-deny + explicit allow: frontend → backend, backend → DB/Redis
ingress/	External traffic entry point
ingress.yaml	Nginx Ingress + cert-manager ClusterIssuer + Let's Encrypt TLS
nginx-configmap.yaml	gzip, TLS 1.2/1.3, JSON logs, rate limit, hidden version
external-secrets/	AWS Secrets Manager → Kubernetes Secrets
external-secrets.yaml	ClusterSecretStore + ExternalSecret (DB creds + Redis creds)
hpa/	Auto-scaling and disruption budgets
hpa-pdb.yaml	HPA frontend(3-15) + backend(3-20) + PodDisruptionBudgets
monitoring/	Observability stack configuration
service-monitors.yaml	ServiceMonitor (backend+frontend) + PrometheusRule alerts
istio-mtls.yaml	PeerAuthentication STRICT + AuthorizationPolicy (allowlist)
grafana-dashboard-cm.yaml	ConfigMap auto-imported by Grafana sidecar (7 panels)

helm/	Helm chart values
values/	Custom values files
prometheus-values.yaml	kube-prometheus-stack: Grafana ingress, 30d retention, alerts
grafana-dashboard.json	7-panel dashboard: RPS, latency, errors, cache, pods, memory

scripts/	Utility and validation scripts
validate.sh	13-point post-deploy check: IPs, secrets, mTLS, TTFB, Prometheus
seed_db.py	Seeds 20 realistic car records into PostgreSQL
local-dev.sh	up / seed / down / migrate / test-api / reset commands

A — Every File's Role

Infrastructure (Terraform)

terraform/providers.tf	—	AWS + K8s + Helm provider auth and S3 state backend
terraform/main.tf	—	Root orchestrator — calls all 8 modules
terraform/variables.tf	—	Centralised config: region, AZs, instance types, CIDRs
terraform/outputs.tf	—	Exports cluster name, ECR URLs, DB/Redis endpoints
modules/vpc/main.tf	—	VPC + 9 subnets + 3 NAT GWs + route tables
modules/eks/main.tf	—	EKS cluster + node group + OIDC + KMS + add-ons
modules/rds/main.tf	—	PostgreSQL 16 Multi-AZ + encrypted + backups
modules/elasticache/main.tf	—	Redis 7 replication group + TLS + Multi-AZ
modules/ecr/main.tf	—	Private image repos + lifecycle rules + deny public
modules/iam/main.tf	—	Least-privilege roles for cluster, nodes, pods (IRSA)
modules/security-groups/main.tf	—	Port-level firewall: 5432/6379 from EKS nodes only
modules/cloudfront/main.tf	—	CDN + 30-day static TTL + private S3 access logs

Containers (Docker)

frontend/Dockerfile	—	3-stage build → 80MB image → non-root uid 1001
frontend/.dockerignore	—	Excludes node_modules, .next, .env from context
backend/Dockerfile	—	2-stage Python build → slim runner → uid 1001
backend/.dockerignore	—	Excludes __pycache__, tests, venv from context
docker-compose.yml	—	Full local stack: Postgres, Redis, backend, frontend

Kubernetes (k8s)

namespaces/namespaces.yaml	—	4 namespaces — frontend + backend get Istio injection
deployments/frontend-deployment.yaml	—	Deployment + Service + SA — zone spread + probes
deployments/backend-deployment.yaml	—	Deployment + Service + SA — IRSA + read-only FS
network-policies/network-policies.yaml	—	Default-deny + allowlist — zero lateral movement
ingress/ingress.yaml	—	Nginx Ingress + cert-manager + Let's Encrypt TLS
ingress/nginx-configmap.yaml	—	gzip + TLS 1.2/1.3 + JSON logs + rate limit
external-secrets/external-secrets.yaml	—	ESO pulls DB+Redis creds from Secrets Manager
hpa/hpa-pdb.yaml	—	HPA (3 → 20 pods) + PDB (min 2 always running)
monitoring/service-monitors.yaml	—	Prometheus scrape config + 4 alerting rules
monitoring/istio-mtls.yaml	—	STRICT mTLS + AuthorizationPolicy allowlist
monitoring/grafana-dashboard-cm.yaml	—	7-panel dashboard auto-imported by Grafana sidecar

CI/CD (GitHub Actions)

.github/workflows/terraform.yml	—	fmt+validate+plan on PR → apply on merge to main
.github/workflows/build-push.yml	—	paths-filter → docker build → Trivy → push ECR
.github/workflows/deploy.yml	—	Helm upgrades → image tag → kubectl apply → rollout
.github/workflows/security-scan.yml	—	Weekly: Trivy+tfsec+Checkov+GitLeaks → SARIF

Scripts & Config

scripts/validate.sh	—	13-point post-deploy security validation script
scripts/seed_db.py	—	Seeds 20 realistic car records into PostgreSQL
scripts/local-dev.sh	—	up / seed / migrate / test-api dev helper
helm/values/prometheus-values.yaml	—	Prometheus+Grafana Helm chart configuration
helm/values/grafana-dashboard.json	—	Grafana dashboard definition (7 panels)
README.md	—	Step-by-step deploy guide + security verification + testing

your-username	feat: production-ready deployment — all security checks passing	2 hours ago	a3f9d12
.github/workflows	Update security scan to run on schedule	3 hours ago	
frontend	Add ContactForm validation and success state	5 hours ago	
backend	Fix Redis TLS connection for ElastiCache	5 hours ago	
terraform	Add CloudFront module and S3 access logs	1 day ago	
k8s	Enforce mTLS STRICT mode in both namespaces	1 day ago	
helm	Add PrometheusRule for cache hit rate alert	2 days ago	
scripts	Update validate.sh with 13 security checks	2 days ago	
.gitignore	Add kubeconfig and *.pem to gitignore	3 days ago	
README.md	Complete deployment guide with verification steps	3 days ago	
docker-compose.yml	Add seed profile for database seeding	4 days ago	

[README.md](#)

[Car Dealership — AWS EKS Production Platform](#)

Full-stack car dealership deployed on AWS EKS with Terraform, Docker, and GitHub Actions.

Architecture

CloudFront CDN → NLB → Nginx Ingress → [Frontend (Next.js) → Backend (FastAPI)] → RDS + Redis

Prerequisites

AWS CLI v2 • Terraform ≥1.6 • kubectl ≥1.29 • Helm ≥3.14 • Docker ≥24

Quick Start (3 commands)

```
$ terraform apply -var="alert_email=ops@you.com"
$ aws eks update-kubeconfig --name car-dealership-prod-eks --region us-east-1
$ ./scripts/validate.sh your-domain.com # 13 security checks
```

[Actions](#) — Recent Workflow Runs

Terraform Plan & Apply
main • a3f9d12 • 2 min 34s • 2h ago
Build & Push to ECR
main • a3f9d12 • 4 min 12s • 2h ago
Deploy to EKS
main • a3f9d12 • 3 min 48s • 2h ago
Security Scan
main • a3f9d12 • 6 min 02s • Monday

[Security](#) — Code Scanning Alerts

0 Critical vulnerabilities	Trivy image scan: frontend + backend	Cleared
0 High vulnerabilities	tfsec: all Terraform modules pass	Cleared
0 Secrets detected	GitLeaks: full git history scanned	Clean
0 Misconfigurations	Checkov: all K8s manifests pass	Cleared